

EUREKA ROADHOUSE, AK, USA

WMO: 702715

Lat: 61.939N

Lon: 147.167W

Elev: 1003

StdP: 89.85

Time Zone: -9.00 (NAK)

Period: 95-19

WBAN: 46407

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.0	(c) -27.4	(d) -33.7	(e) 0.2	(f) -25.5	(g) -32.6	(h) 0.2	(i) -25.5	(j) 8.4	(k) -8.6	(l) 7.4	(m) -8.8	(n) 2.1	(o) 50	(p) 0.818

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.3	(c) 20.8	(d) 12.1	(e) 18.0	(f) 10.9	(g) 17.0	(h) 10.2	(i) 12.9	(j) 18.5	(k) 11.8	(l) 16.7	(m) 10.8	(n) 15.3	(o) 2.7	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 10.2	(b) 8.7	(c) 14.2	(d) 9.0	(e) 8.1	(f) 13.1	(g) 8.0	(h) 7.5	(i) 12.1	(j) 39.3	(k) 19.0	(l) 36.2	(m) 16.8	(n) 33.5	(o) 15.3	(p) 17.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	6.4	5.5	-31.4	23.5	1.8	2.2	-32.7	25.1	-33.7	26.3	-34.8	27.6	-36.1	29.1
			-31.0	13.9	2.0	1.6	-32.4	15.0	-33.6	16.0	-34.7	16.9	-36.1	18.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.1	-16.1	-13.5	-11.5	-3.6	4.0	9.5	11.1	9.5	4.3	-4.0	-12.7	-15.0		
	DBStd	11.10	6.72	6.21	5.72	4.41	3.92	3.24	2.54	2.89	3.35	4.40	5.73	5.90		
	HDD10.0	4902	808	658	665	408	191	48	17	44	173	434	681	774		
	HDD18.3	7828	1067	891	923	658	445	267	224	274	422	693	931	1033		
	CDD10.0	117	0	0	0	0	4	32	52	28	1	0	0	0		
	CDD18.3	1	0	0	0	0	0	1	0	0	0	0	0	0		
	CDH23.3	6	0	0	0	0	0	5	1	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	2.2	1.6	1.7	2.1	2.3	2.9	3.2	3.0	2.6	2.3	1.6	1.5	1.4		
(15) Precipitation	PrecAvg	1282	95	104	78	68	69	75	108	135	165	141	105	121		
	PrecMax	1610	283	270	153	135	159	119	175	360	288	354	295	240		
	PrecMin	867	5	0	9	15	20	21	48	25	37	63	1	19		
	PrecStd	165	67	59	50	30	39	23	35	78	69	76	77	64		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	-0.9	2.5	9.1	18.7	23.6	22.2	22.6	15.0	8.7	1.1	-0.1		
		MCWB	-1.4	-2.6	-0.1	4.0	9.4	12.8	13.2	13.6	8.4	5.0	-0.7	-1.0		
	2%	DB	-2.6	-2.3	0.0	6.1	16.2	20.6	19.9	19.0	12.4	4.1	-1.2	-2.5		
		MCWB	-3.6	-3.4	-1.9	1.4	7.5	11.4	12.2	12.0	6.9	1.8	-1.7	-3.3		
	5%	DB	-5.0	-3.7	-2.3	3.8	12.9	17.7	17.8	17.0	11.0	2.6	-2.7	-4.8		
		MCWB	-5.9	-5.1	-3.9	0.2	6.2	10.2	11.3	10.9	6.3	0.6	-3.4	-5.6		
	10%	DB	-7.3	-6.0	-3.8	2.5	11.2	15.9	16.4	14.0	8.9	1.3	-4.8	-7.1		
		MCWB	-7.8	-6.7	-5.6	-0.7	5.2	9.1	10.4	9.4	5.4	-0.3	-5.7	-7.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	-1.3	0.1	4.1	9.6	14.6	14.0	14.2	8.8	5.4	-0.4	-0.9		
		MCDB	-0.3	-0.4	2.3	8.7	17.5	21.0	20.0	21.0	13.5	8.2	0.5	-0.6		
	2%	WB	-3.4	-3.2	-1.8	2.0	7.9	12.3	12.9	12.7	7.4	2.6	-2.0	-3.2		
		MCDB	-2.5	-2.4	-0.3	5.7	15.3	18.2	18.3	18.0	11.2	3.9	-1.5	-2.7		
	5%	WB	-5.7	-4.6	-3.6	0.6	6.6	10.7	11.9	11.3	6.6	0.8	-3.4	-5.4		
		MCDB	-4.9	-3.1	-2.3	3.8	12.6	16.3	16.7	15.6	9.8	2.1	-2.6	-4.7		
	10%	WB	-7.6	-6.3	-5.3	-0.4	5.3	9.5	11.1	10.2	5.7	-0.1	-5.4	-7.2		
		MCDB	-7.1	-5.5	-3.7	2.3	10.5	14.7	15.6	13.7	8.6	1.2	-4.6	-6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.0	7.0	7.5	8.6	9.1	8.3	7.6	6.6	5.1	5.4	5.9		
		MCDBR	6.6	7.1	7.5	7.9	12.0	11.9	10.6	10.3	8.6	6.3	5.7	6.9		
		MCWBR	6.2	6.3	6.1	5.5	6.3	5.5	5.1	5.1	4.9	4.4	5.2	6.7		
	5% WB	MCDBR	6.7	6.8	7.3	7.6	11.4	11.0	9.7	9.7	7.9	5.5	5.5	6.8		
		MCWBR	6.4	6.2	5.9	5.2	6.1	5.2	4.8	5.0	4.7	4.1	5.2	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.213	0.279	0.285	0.334	0.366	0.355	0.357	0.354	0.306	0.273	0.224	0.202		
	taud		1.983	2.030	2.102	2.024	2.147	2.315	2.333	2.343	2.435	2.232	1.939	1.838		
	Ebn at Noon		473	657	819	842	847	868	855	825	810	679	421	259		
	Edh at Noon		38	76	101	136	131	114	109	99	74	60	38	24		
(44) All-Sky Solar Radiation	RadAvg		0.29	1.07	2.83	4.79	5.62	5.64	5.13	3.96	2.54	1.18	0.42	0.14		
	RadStd		0.03	0.10	0.19	0.26	0.34	0.31	0.24	0.26	0.23	0.11	0.05	0.01		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (21 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page